

Global Renewable Energy in 2025:

How Far Have We Come?

*A data-driven overview of renewable energy's growing share
in global primary energy and electricity markets.*

Audience: Policy analysts & energy sector strategists

75% of Global GHG Emissions Still Come from Fossil Fuels

The case for accelerating the energy transition

The Scale of the Problem

Since the Industrial Revolution, fossil fuels have dominated energy systems worldwide. Burning coal, oil, and gas for energy produces three-quarters of all greenhouse gas emissions.

75%

of global GHG emissions
from fossil fuel
combustion

5M+

premature deaths/year
from air pollution

196+

nations committed to
net-zero by 2050

The energy transition must rapidly scale low-carbon sources — nuclear and renewable technologies — to bend the emissions curve.

Why Renewables Are Critical

- Fossil fuel combustion is the #1 driver of climate change
- Air pollution causes at least 5 million premature deaths/year
- Energy systems must decarbonize to meet Paris Agreement goals
- Renewables (solar, wind, hydro, geothermal, bioenergy, wave, tidal) are the primary alternative
- Traditional biomass excluded from modern renewable statistics (often significant in lower-income settings)

Renewables Now Supply ~14% of Global Primary Energy

Approximately one-seventh of the world's primary energy comes from renewable technologies (2024, substitution method)

What "Primary Energy" Means

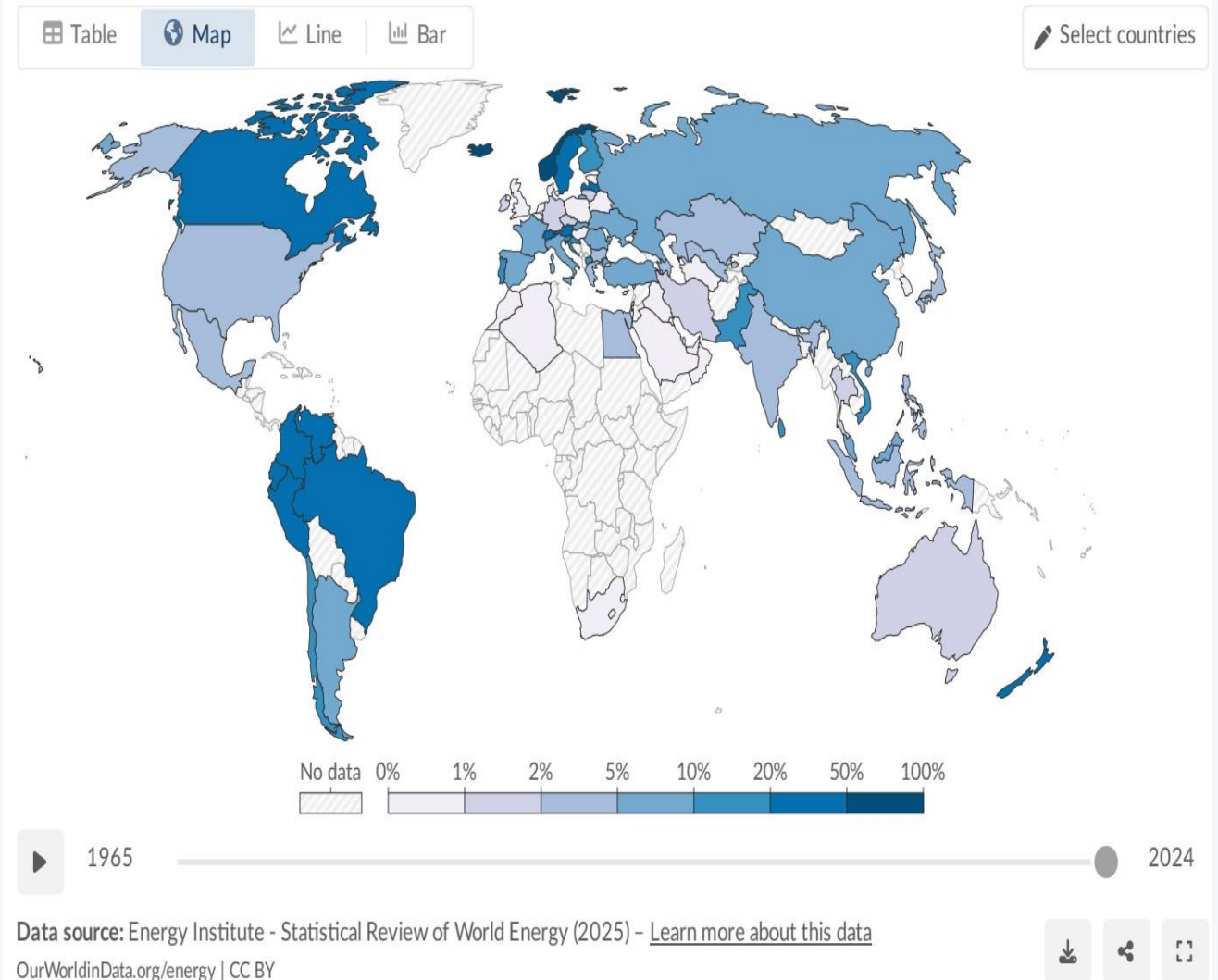
Primary energy encompasses all energy sectors combined:

- Electricity generation
- Transport (petrol, diesel, jet fuel)
- Heating (natural gas, coal, oil)

Because transport and heating are harder to decarbonize (still heavily reliant on oil and gas), the renewable share in the total energy mix is lower than in electricity alone.

Key Insight

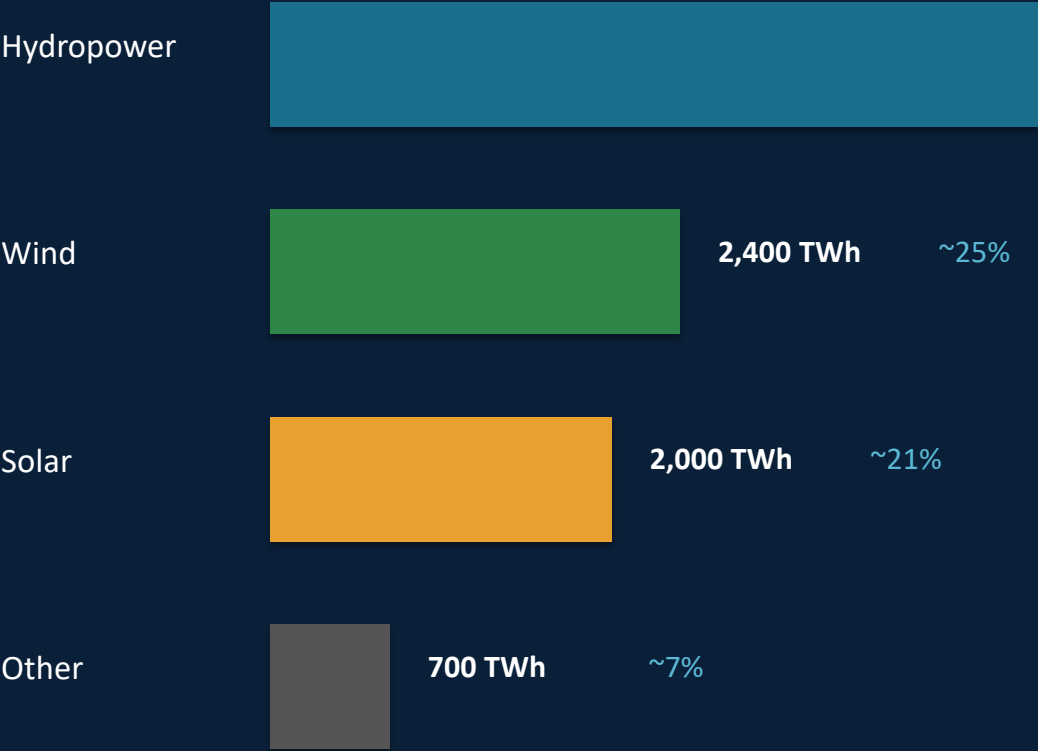
Fossil-fuel-producing nations in the Middle East and parts of Asia remain below 5% renewable in primary energy. Countries like Norway, Brazil, and New Zealand derive the largest shares from renewables.



Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Global renewable electricity generation by technology — hydro, wind, solar, and other

Global Renewable Generation by Technology (2024)



Wind + solar combined now nearly match hydropower's output

Technology Breakdown (2024 Data)

Technology	Generation	Share
Hydropower	~4,500 TWh	~47%
Wind	~2,400 TWh	~25%
Solar	~2,000 TWh	~21%
Other	~700 TWh	~7%
TOTAL	~9,600 TWh	100%

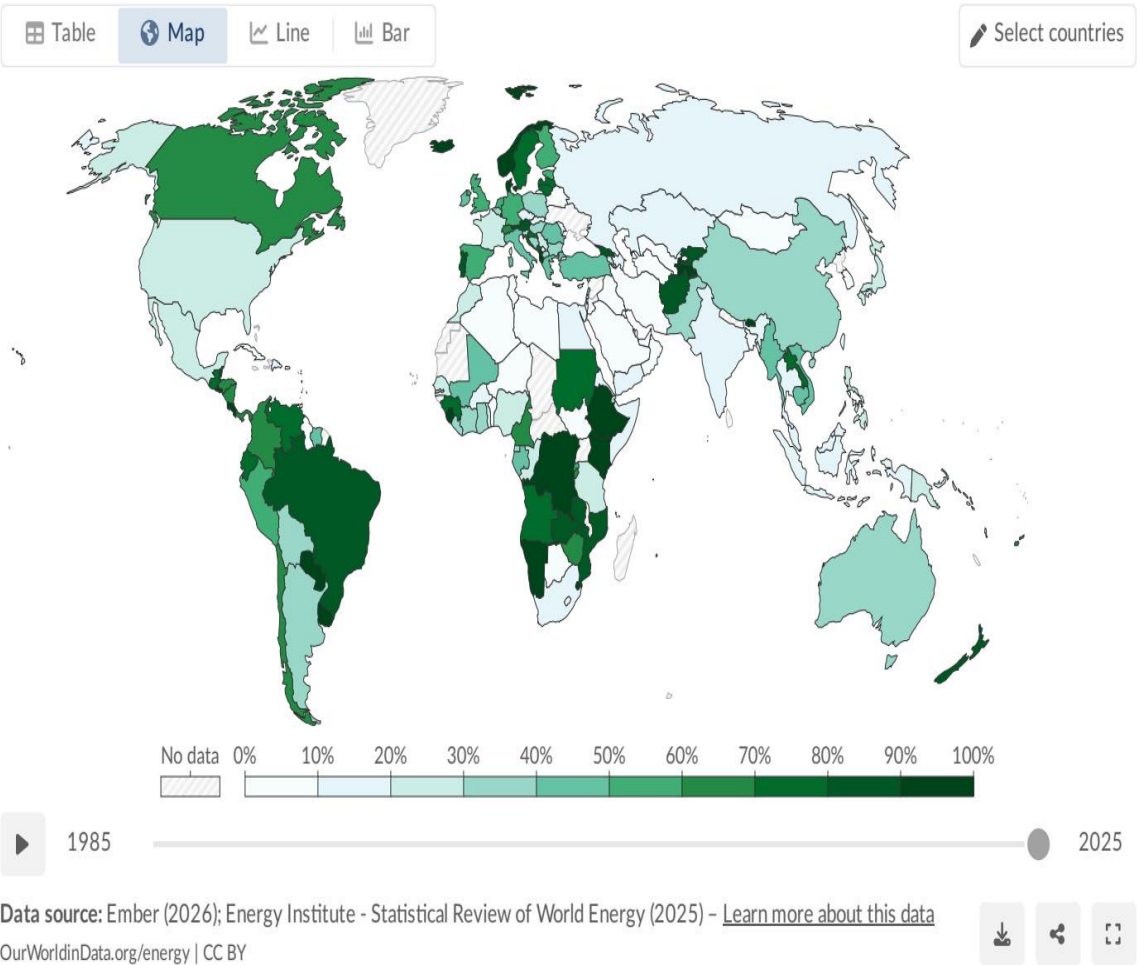
Wind and solar have accelerated dramatically since ~2010, driven by falling costs and policy support worldwide.

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity from renewables by country — 2025 data

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.



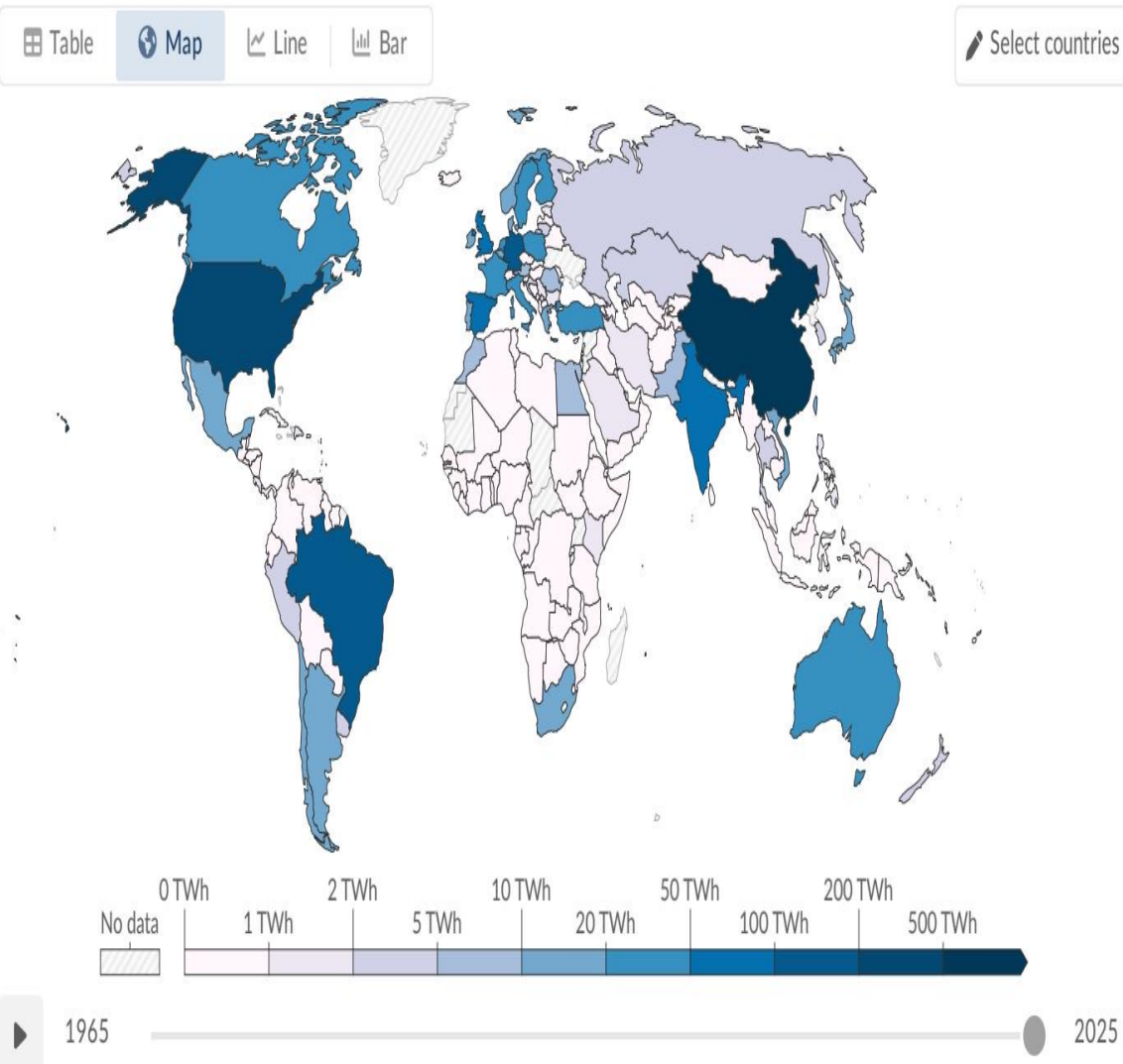
Selected Countries & Regions

Renewable Share of Electricity (2025)

Country / Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
Germany	~55%
China	~35%
United States	~25%
India	~25%
Saudi Arabia	<2%
World Average ≈ 30% Middle East & parts of Asia: <10%	

Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower has been generating electricity at scale for over a century — top producers: China, Brazil, Canada



~4,500 TWh

Global hydropower generation (2024)

~47%

of all renewable electricity

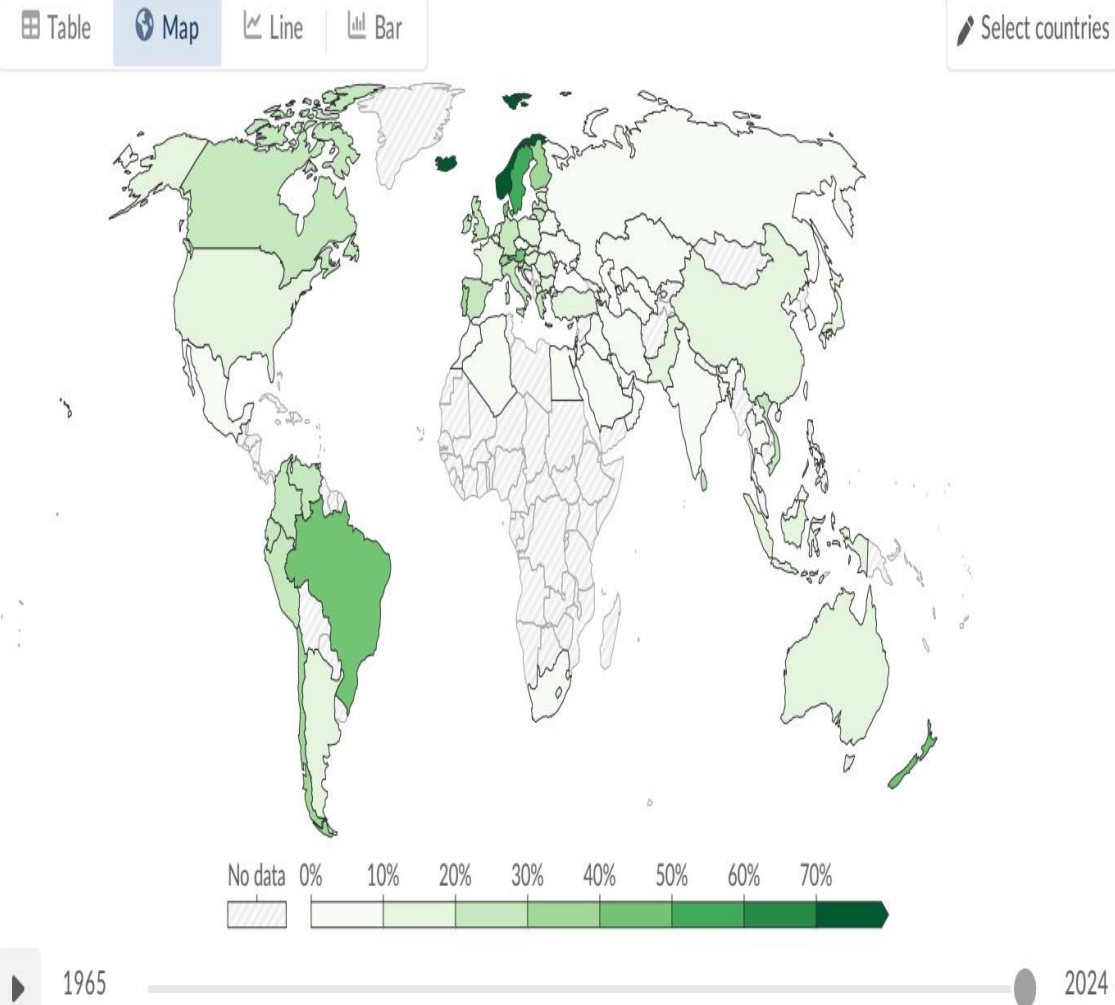
Key Facts

- First deployed at commercial scale over a century ago
- Accounts for roughly half of all renewable electricity worldwide
- Top producers: China (~1,300 TWh), Brazil (~650 TWh), Canada (~400 TWh)
- Limited growth potential vs. wind and solar — mature technology
- Run-of-river, reservoir, and pumped hydro installations
- Provides baseload power and grid stability services

Wind and Solar Are the Fastest-Growing Energy Technologies in History

Wind and solar combined now nearly match hydropower's output — solar is the fastest-growing source

Wind Power Generation by Country (2025)



Wind vs. Solar: Generation Comparison (2024)

WIND

~2,400 TWh

≈ 25% of renewable
electricity

SOLAR

~2,000 TWh

≈ 21% of renewable
electricity

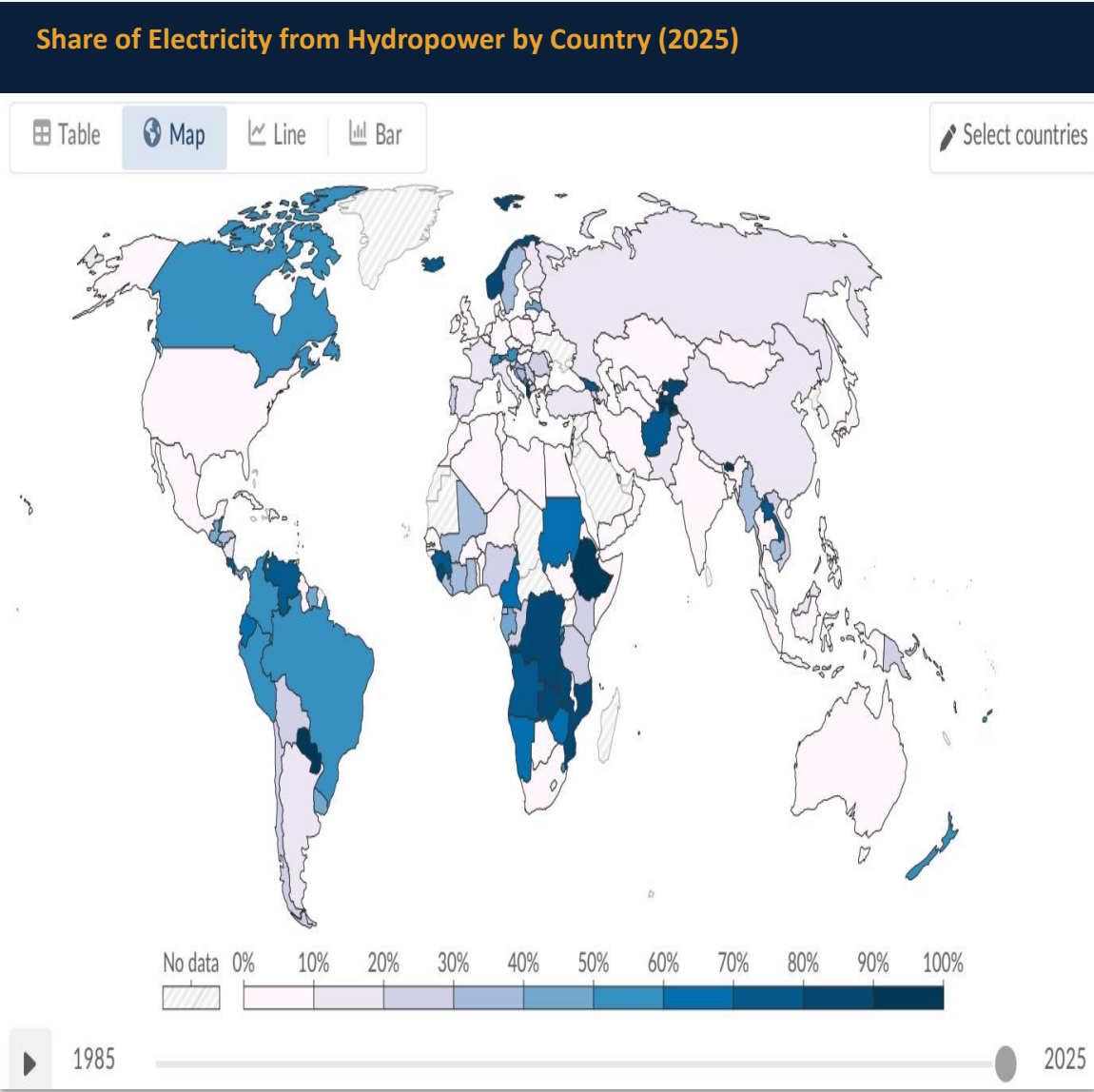
Wind + Solar combined ≈ 4,400 TWh — nearly matching hydropower's ~4,500 TWh

Solar: The Fastest-Growing Source

- Global solar generation has increased dramatically since 2010
- Driven by rapidly falling costs and strong policy support worldwide
- Growth trajectory now outpacing all other energy technologies in history
- Major producers: China, EU, US, India

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead

The energy transition is highly uneven — hydropower access and renewable investment vary dramatically by region



Regional Disparities in the Energy Transition

Uneven investment in renewables, fossil fuel lock-in, and infrastructure gaps mean the transition is far from uniform.

Key Takeaways

Renewables are growing fast — but the transition must accelerate

~14%

of global primary energy

now comes from renewable sources — a milestone, but far from sufficient for climate targets

~30%

of global electricity

now comes from renewables — electricity is where the transition is fastest

Hydropower

still dominates

at ~47% of renewable generation, but its growth potential is limited vs. wind & solar

Wind + Solar

are the fastest-growing

technologies in history — combined, they nearly match hydropower's output

Regional gaps

are vast

Scandinavia & Latin America lead at 60–98%; Sub-Saharan Africa & Middle East remain below 10%

Transport & heating

lag far behind

keeping the primary energy share at only ~14% vs. ~30% for electricity

Accelerating the transition across all sectors — electricity, transport, and heating — is essential to meet Paris Agreement goals and limit global warming to 1.5°C.